Lab 7: Fuzzing for Input Validation Bugs - PES1UG24CS498

Step 2: Output of the original (buggy) processor.py

Command: python processor.py

```
PS C:\Users\TEJAS PONNAPPA\Downloads\lab7_buggy> python processor.py
==== Running parse_transit_corridor ====
Enter corridor (e.g. DT-UP-3): bad
Traceback (most recent call last):
  File "C:\Users\TEJAS PONNAPPA\Downloads\lab7_buggy\processor.py", line 39, in <module>
    main()
    ~~~~^^
  File "C:\Users\TEJAS PONNAPPA\Downloads\lab7_buggy\processor.py", line 28, in main
    print(parse_transit_corridor(c_input))
          ~~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^
  File "C:\Users\TEJAS PONNAPPA\Downloads\lab7_buggy\processor.py", line 2, in parse_transit_corridor
    orig, dest, zones_str = corridor_code.strip().split('-')
    ^^^^^^^^^^^^^^^^^^^^^
ValueError: not enough values to unpack (expected 3, got 1)
```

Step 4: pytest against the buggy processor (tests failing)

Command: python -m pytest test_processor.py

```
C:\Users\TEJAS PONNAPPA\Downloads\lab7_buggy> python -m pytest test_processor.py
 +---------------- 3 ----------------
 | Traceback (most recent call last):
 |   File "C:\Users\TEJAS PONNAPPA\Downloads\lab7_buggy\test_processor.py", line 21, in test_compute_passenger_density_safe
 |     compute_passenger_density(s)
 |     ~~~~~~~~~~~~~~~~~~~~~~~~~^^^
 |   File "C:\Users\TEJAS PONNAPPA\Downloads\lab7_buggy\processor.py", line 19, in compute_passenger_density
 |     passengers_str, km_str = occupancy_ratio_str.split('/')
 |                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^
 | AttributeError: 'NoneType' object has no attribute 'split'
 |
 | During handling of the above exception, another exception occurred:
 |
 | Traceback (most recent call last):
 |   File "C:\Users\TEJAS PONNAPPA\Downloads\lab7_buggy\test_processor.py", line 23, in test_compute_passenger_density_safe
 |     assert False, f"compute_passenger_density crashed with: {e!r}"
 | AssertionError: compute_passenger_density crashed with: AttributeError("'NoneType' object has no attribute 'split'")
 | assert False
 | Failing test case: test_compute_passenger_density_safe(
 |     s=None,  # or any other generated value
 | )
 +-----------------------------------
```

```
=========================================================== short test summary info ============================================================
FAILED test_processor.py::test_parse_transit_corridor_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_processor.py::test_lookup_concession_discount_safe - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_processor.py::test_compute_passenger_density_safe - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
============================================================ 3 failed in 3.49s =============================================================
```

Step 5: Output of the fixed processor.py

Command: python processor.py

```
==== Running parse_transit_corridor ====
Enter corridor (e.g. DT-UP-3): bad
Invalid corridor code

==== Running lookup_concession_discount ====
Enter concession category (e.g. STUDENT, SENIOR): student
Multiplier: 0.5

==== Running compute_passenger_density ====
Enter passengers/km (e.g. 50/10.0): 50/10.0
Density: 5.0
```

Step 6: pytest against the fixed processor (tests passing)

Command: `python -m pytest test_processor.py`

```
C:\Users\TEJAS PONNAPPA\Desktop\SE Lab\L6_L7_10_bus_transit_fare\Lab 7> python -m pytest test_processor.py
============================================================ test session starts ============================================================
tform win32 -- Python 3.13.7, pytest-9.1.1, pluggy-1.7.0
tdir: C:\Users\TEJAS PONNAPPA\Desktop\SE Lab\L6_L7_10_bus_transit_fare\Lab 7
gins: anyio-4.12.1, Faker-37.11.0, hypothesis-6.170.15, langsmith-0.14.13, django-4.14.0
lected 3 items

t_processor.py ...                                                                                                                  [100%]

============================================================ 3 passed in 0.71s =============================================================
```